

skyjam — Forecasting GNSS Interference over European Airspace

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Repository (public): <https://github.com/Zayden16/hslu-mlops-hs-26>

1 Problem statement

What and for whom. GNSS jamming and spoofing are now routine over the Baltic, the Black Sea and the Eastern Mediterranean: aircraft lose the integrity of their satellite position solution mid-flight. Public tools such as gpsjam.org render *yesterday's* map. A dispatcher planning a route, a drone operator or a flight-ops centre needs the opposite, a statement about the hours ahead.

Prediction. For every H3 resolution-2 airspace cell (~180 km edge), predict the probability that the cell will be *interference-affected exactly 6 hours ahead* (a binary classification per cell-hour, produced every hour).

Scope. European and adjacent airspace, covered by 18 sampling discs of 250 NM: 11 over regions where interference is reported, 7 over Western-European control regions. One hourly sweep currently yields 2 272 cruise-level observations of 1 537 distinct aircraft over 67 populated cells.

Label. A cell-hour is affected when at least **30 %** of the cruise-altitude aircraft in it broadcast a degraded navigation integrity category (NIC < 7, i.e. a containment radius worse than 0.2 NM). NIC is each aircraft's own assessment of its position quality, transmitted over ADS-B. The label is therefore a *fleet-level consensus about a physical event*, aggregated over independent airframes at a future timestamp, not a rule applied to the model's inputs.

Success criterion. On a strictly out-of-time test split, the model must beat two baselines:

Baseline	Why it is the bar
Persistence: state at t repeated at $t + 6$ h	Interference is strongly autocorrelated; not beating this means nothing was learned
Per-cell hourly climatology	Encodes “the Gulf of Finland is usually bad at 14:00” with no live signal

Target: PR-AUC at least 0.10 above persistence, and a lower Brier score, with the gain holding *separately* in the interference regions and the control regions. The positive class is rare (4.5 % of cells in the validation snapshot, 3/67), so accuracy is meaningless and PR-AUC is the headline metric. It is handled with class weighting, a decision threshold tuned on the validation split, and precision/recall reported per region, because a model that permanently flags the whole Baltic would look good globally and be operationally useless.

2 Originality & motivation

Checked against the HSLU archive (mlops-lab.ch) and the KTH ID2223 project lists: both are dominated by crypto prices, weather, air quality and transport availability. Nothing there predicts the *quality* of a navigation signal, and no ADS-B project appears.

In one sentence: the dataset does not exist until my pipeline creates it, and the label is inferred from fleet telemetry rather than read from a column. Concretely:

1. **No historical download exists** for per-cell navigation integrity. The feature pipeline is the sole source of history, which makes the scheduled job load-bearing rather than decorative.
2. **The label is constructed**, from a fleet-level statistic over self-reported avionics quality, with a defensible threshold and an explicit traffic floor.
3. **Drift is the subject, not an afterthought.** A new emitter switching on is genuine concept drift in the region it covers, so the MS4 monitoring measures the very phenomenon being modelled.

Motivation: GNSS interference is an active civil-aviation safety concern in Europe, the raw telemetry is fully public, and I want a project whose failure modes are physical rather than financial.

3 Data source & features

Source. `api.adsb.lol` v2, a public community ADS-B aggregator, endpoint `/v2/point/{lat}/{lon}/{radius}`. It is a documented JSON API, **not scraping**, free and key-less for non-commercial use. **Fallback:** `api.adsb.fi` serves the same schema and is a one-line configuration change (`SKYJAM_ADSB_BASE_URL`).

Update frequency and volume. The feed is real time; the pipeline polls all 18 points **hourly** with paced requests and exponential backoff on HTTP 429. History **starts when the poller starts**: there is no back-download. At ~67 cell-rows per sweep the store grows by roughly **1 600 cell-hours per day**, so MS3 in December will have on the order of 100 000 labelled rows. Ingestion is already running to accumulate history ahead of MS2.

Validated signal (live snapshot, aircraft at or above FL200, degraded = NIC < 7):

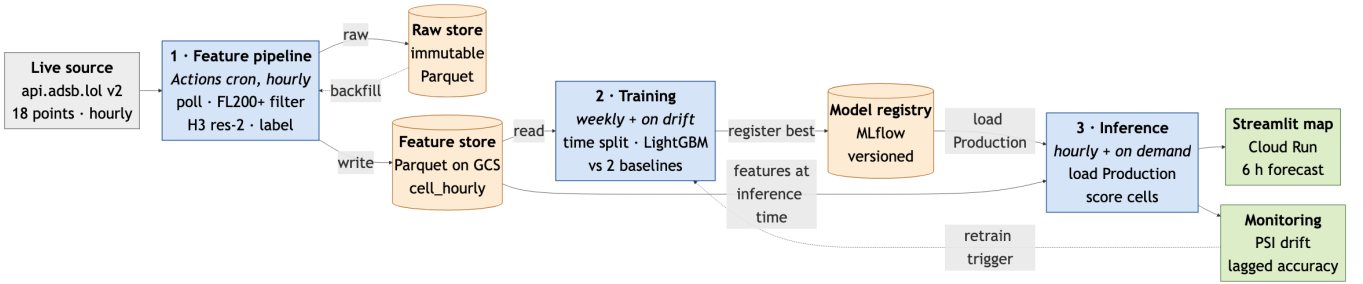


Figure 1: FTI architecture. Solid arrows are the core data flow; dotted arrows are the backfill and retrain triggers.

Interference region		Interference Deg. region		Control region	
					Deg.
Gulf of Finland	52.0 %	Baltic south	13.2 %	Germany	2.4 %
Poland east	34.0 %	Romania/Moldova	6.2 %	Switzerland	1.4 %
Kaliningrad	25.9 %	Black Sea west	5.1 %	UK / France	1.0 / 0.0 %

A ~40-fold separation between the worst region and the controls, so the target is real rather than assumed. **The altitude filter is a feature, not a formality:** an unfiltered probe of the Swiss disc shows 8.9 % degraded aircraft against 2.3 % at FL200+, because cheap general-aviation avionics report poor integrity everywhere; the same filter leaves the Baltic figures essentially untouched. Regional shares move hour to hour, which is the phenomenon being forecast, not noise in the measurement.

Named features, per cell and hourly slot: `aircraft_count`, `degraded_share`, `median_nic`, `p25_nic`, `mean_nac_p`; lags of `degraded_share` and of the binary state at 1, 2, 3, 6, 12, 24 and 168 h; backward-looking rolling means of `degraded_share` over 6 h and 24 h; `aircraft_count_mean_24h`; and the calendar features `hour_utc` and `day_of_week`.

Leakage. (i) *Split*: strictly chronological — train on the earliest period, validate on the middle, test on the most recent, with a 6-hour gap between segments so no example straddles a boundary. No shuffling anywhere. (ii) *Temporal, not positional lags*: lags are computed after reindexing each cell onto a gap-free hourly grid, because a positional `shift(1)` would silently present a three-hour-old observation as one hour old whenever the cron skipped a run; a unit test asserts the lag across a deliberately removed hour is `NaN`. (iii) *Rolling windows exclude the present*, shifted by one slot before aggregation, verified by test. (iv) *Unobserved is not quiet*: cells below five aircraft are dropped rather than imputed as clean, and rows whose future state was never observed carry no label.

4 System design

Pipelines and triggers. (1) *Feature*: hourly GitHub Actions cron polls all 18 points, aggregates to cell-hours and appends to the store; raw snapshots are retained immutably so the derived layer can be rebuilt end to end when a definition changes, which is exactly what `skyjam-backfill` does. (2) *Training*: weekly and on demand, reads the feature store, builds the supervised table, splits chronologically, trains against both baselines, logs to MLflow and registers the best run. (3) *Inference*: hourly and on UI request, loads the Production model and writes a 6-hour forecast per cell, served as a map.

Stack, one line each:

- **Feature store** — versioned Parquet on GCS, date-partitioned: point-in-time correct and append-only; a managed store would add cost and no capability at this scale.
- **Orchestration** — GitHub Actions cron: the scheduler lives next to the code, and since a missed hour is permanently lost data, simplicity *is* reliability.
- **Tracking / registry** — MLflow: run comparison plus a Production alias for inference to load, so the deployed version is always identifiable.
- **Serving** — Streamlit map on Google Cloud Run: scales to zero within the USD 50 credit, and a map is the natural view of a spatial forecast.
- **Packaging** — Docker + `uv.lock`: one identical image in CI, cron and Cloud Run.

Training–serving skew is prevented structurally: the thresholds, the altitude floor, the H3 resolution and the lag set live in `src/skyjam/common/schema.py`, imported by all three pipelines, and inference reuses the feature pipeline’s own aggregation function.

Optional / stretch (explicitly not core): Terraform IaC, Airflow, a managed feature store, and alerting on the drift metric. Core FTI comes first.

Current state. The feature pipeline, aggregation, leakage logic, backfill, Dockerfile and CI are implemented and green (24 unit tests, written against deliberately introduced leakage bugs). **The repository is public:** <https://github.com/Zayden16/hslu-mlops-hs-26>

Risks. History accrues only from now on — mitigated by starting ingestion at proposal time and retaining raw immutably so features can be recomputed. The public API could rate-limit or disappear — pacing, backoff and the `adsb.fi` fallback. Sparse night traffic — a per-cell traffic floor, with unobserved cells excluded rather than imputed.